



Novel M (Mg/Ni/Cu)-Al-CO₃ layered double hydroxides synthesized by aqueous miscible organic solvent treatment (AMOST) method for CO₂ capture

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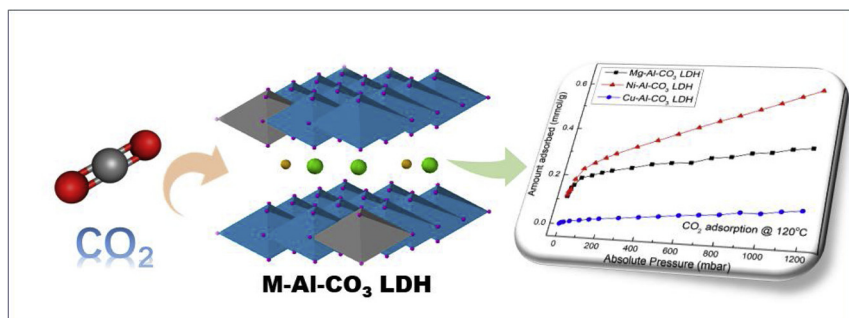
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GRAPHICAL ABSTRACT



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ABSTRACT

Layered double hydroxides (LDHs) have been intensively studied in recent years owing to their great potential in CO₂ capture. However, the severe aggregation between platelets and low surface area restricted it from exhibiting very high CO₂ adsorption capacity and CO₂/N₂ selectivity. In this research, we for the first time synthesized Ni-Al-CO₃ and Cu-Al-CO₃ LDHs using aqueous miscible organic solvent treatment (AMOST) method. The as-synthesized materials were evaluated for CO₂ adsorption at three different temperatures (50, 80, 120 °C) applicable to post-combustion CO₂ capture. Characterized with XRD, N₂ adsorption-desorption, TEM, EDX, and TGA, we found the newly synthesized Ni-Al-CO₃ LDH showed a nano-flower-like morphology comprising randomly oriented 2D nanoplatelets with both high surface area (249.45 m²/g) and pore volume (0.59 cc/g). Experimental results demonstrated that un-calcined Ni-Al-CO₃ LDH is superior in terms of CO₂ capture among the three LDHs, with a maximum CO₂ adsorption capacity of 0.87 mmol/g and the ideal CO₂/N₂ selectivity of 166 at 50 °C under 1200 mbar for typical flue gas CO₂/N₂ composition (CO₂:N₂ = 15:85, v/v). This is the first

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report of a delaminated Ni-Al-CO₃ LDH showing better CO₂ capture performance than the well-reported optimal Mg layered double hydroxide.

1. Introduction

With the acceleration of industrialization, the rapid development of modern society has been accompanied with the huge consumption of fossil fuels, leading to the emission of large amount of greenhouse gases, especially, carbon dioxide (CO₂) [1]. The atmospheric concentration of CO₂ increased from its pre-industrial level of 280 ppm to 380 ppm in 2005, and is expected to reach 550 ppm by 2050 if CO₂ emission continues at the same rate in the next three decades [2,3]. This exponentially increasing anthropogenic emission of CO₂ has already resulted in the rise of average global temperature, a phenomenon referred to as global warming. The temperature due to global warming is predicted to increase by 1.1–6.4 °C during the 21st century [3], threatening the survivals of both humans and animals on the earth. Therefore, there is an urgent need to take measures to reduce the carbon emission. CO₂ capture is viewed as one of the potential strategies for immediate action towards climate change mitigation. Technologies attempted for CO₂ capture are mainly based on solvent absorption [4], adsorption on solid adsorbents [5] and membrane separation [6]. Among these technologies, the use of solid adsorbents for CO₂ capture is deemed to be more economical and simpler than the technologically mature solvent absorption process. The choice of adsorbents plays a central role in deciding the overall efficiency of adsorption technology for CO₂ selective separation and capture. Many types of solid adsorbents, such as activated carbon [7,8], zeolites [9–16], and metal-organic frameworks [17–21] etc., have been exploited for CO₂ capture. However, these adsorbents usually show low adsorption efficiency at a relatively high temperature for flue gas, which is usually above 100 °C. Therefore, it is important to develop high performance adsorbents capable of working at such high temperatures for CO₂ capture.

Layered double hydroxides (LDHs), also known as hydrotalcite-like compounds (HTs) or anionic clays, are a kind of ionic lamellar compounds consisting of positively charged brucite-like layers with an interlayer region containing charge compensating anions and solvation molecules [22]. A generic formula of most studied LDHs can be written as $[M_1^{2+}_x M_2^{3+}_{1-x}(\text{OH})_2][A^{n-}]_x \cdot n\text{H}_2\text{O}$, where M^{2+} and M^{3+} are divalent (e.g., Mg²⁺, Zn²⁺, Ni²⁺) and trivalent cations (e.g., Al³⁺, Ga³⁺, Fe³⁺, Mn³⁺), respectively. A^{n-} is a non-framework charge compensating anion (e.g., CO₃²⁻, Cl⁻, SO₄²⁻), and x is usually in the range of 0.2–0.4 [22–24]. LDHs have found a wide range of applications, such as being used as CO₂ adsorbents [25–27], heavy metal ions adsorbents [28,29], catalysts [30–32], and drug delivery hosts [33]. Among various LDHs, the most widely studied one for CO₂ capture is Mg-Al-CO₃ LDH as well as its derived materials because LDHs would transform into mixed metal oxides (MMOs) upon high temperature calcination, which could provide active adsorption sites Mg-O for CO₂ [3,27], thus increasing CO₂ adsorption capacity. It is generally acknowledged that the fresh LDH has no CO₂ adsorption ability [3].

Due to the great potential of LDHs as CO₂ adsorbents, researchers developed several synthesis methods for LDHs with the aim of optimizing these materials. The commonly used one is co-precipitation method. This method involves simultaneous addition of divalent and trivalent metal salts solution into a base and interlayer anion solution (e.g., NaOH, Na₂CO₃) at a constant pH (e.g., 10) to allow both metal salts to co-precipitate under the same pH conditions. LDHs synthesized through this process typically show severe aggregation with *ab*-face stacking, resulting in large-sized crystal particles and low surface areas, which leads to underutilization of adsorption sites of the adsorbent. In order to solve this problem, O'Hare and co-workers [34] managed to

optimize synthetic procedures using aqueous miscible organic solvent treatment (AMOST) method to successfully obtain Zn₂Al-borate and Mg₃Al-borate LDHs containing delaminated nanosheets with a uniform particle size of ca. 5 μm and high specific surface areas of 458.6 and 263 m² g⁻¹, respectively. Regarding the AMOST method, the LDHs are initially formed using a conventional coprecipitation approach but before final isolation the solid is re-dispersed in an aqueous miscible organic solvent, such as methanol or acetone, which can change the structure of LDHs during redispersion process. Buffet et al. [35] subsequently reported methylaluminoxane modified AMOST Mg₆Al₂(OH)₁₆CO₃·4H₂O LDH and used it as a catalyst support for the slurry phase polymerisation of ethylene. Chen et al. [36] also adopted AMOST method to tune the surface area and particle morphology of Mg/Al-CO₃ LDH by adjusting the organic solvent amount, re-dispersion time, and acetone washing steps.

The above-mentioned studies show that the modifications during the synthesis of LDH with AMOST are effective in improving the morphology, dispersity, surface area, and pore volume. In this context, we hypothesized that the optimized LDH could show large CO₂ adsorption capacity and high CO₂/N₂ selectivity. Herein, we synthesized three types of M (Mg/ Ni/ Cu)-Al-CO₃ layered double hydroxides using AMOST method and investigated their CO₂ adsorption performance at three different temperatures (50, 80, 120 °C) applicable to post-combustion CO₂ capture. We found that the novel Ni-Al-CO₃ LDH displayed the best CO₂ capture ability at all three temperatures among the three LDHs. This is the first report of an un-calcined Ni-Al-CO₃ LDH possessing the superior CO₂ adsorption ability to the nominal best LDH-based CO₂ adsorbent, i.e., Mg-Al-CO₃ LDH. This newly synthesized Ni-Al-CO₃ LDH using AMOST method may afford a good candidate as CO₂ adsorbent and also provide a route for tuning other LDH materials in a wide range of applications.

2. Experimental

2.1. Materials

Mg(NO₃)₂·6H₂O, Ni(NO₃)₂·6H₂O, Cu(NO₃)₂·6H₂O, Al(NO₃)₃·9H₂O, and Na₂CO₃ with AR grade were all purchased from Alfa Aesar. NaOH and acetone were purchased from Sigma-Aldrich. Milli-Q water was used throughout the experimental process. All chemicals were used without further purification.

2.2. Synthesis of M-Al-CO₃ layered double hydroxides

In this study, we used the aqueous miscible organic solvent treatment (AMOST) method [36] with some modifications to synthesize three M-Al-CO₃ LDHs instead of the conventional co-precipitation method. This is the first time that Ni and Cu LDHs are synthesized using AMOST method. The schematic structural of three M-Al-CO₃ LDHs can be seen in Fig. 1.

For the synthesis of Mg-Al-CO₃ LDH, firstly, 37.5 mmol (9.6 g) of Mg(NO₃)₂·6H₂O and 12.5 mmol (4.78 g) of Al(NO₃)₃·9H₂O were dissolved in 20 ml Milli-Q water by magnetic stirring as Solution A. 25 mmol (2.65 g) of Na₂CO₃ was dissolved in 50 ml of Milli-Q water by magnetic stirring at room temperature as Solution B. 200 mmol of NaOH (8 g) was dissolved in 50 ml of Milli-Q water as Solution C. Then, Solution A was added drop-wise into Solution B with continuous stirring. At the same time, the solution pH value was kept around 9–10 by adding Solution C to the above mixture. After Solution A was added completely to Solution B, the mixture was stirred for another 30 min at a stirring

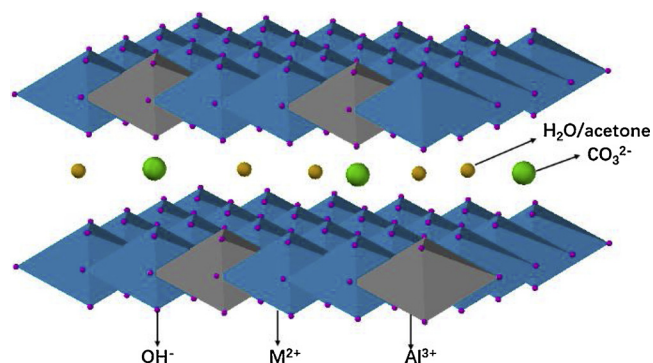


Fig. 1. Schematic structural representation of M-Al-CO₃ LDHs synthesized by AMOST.

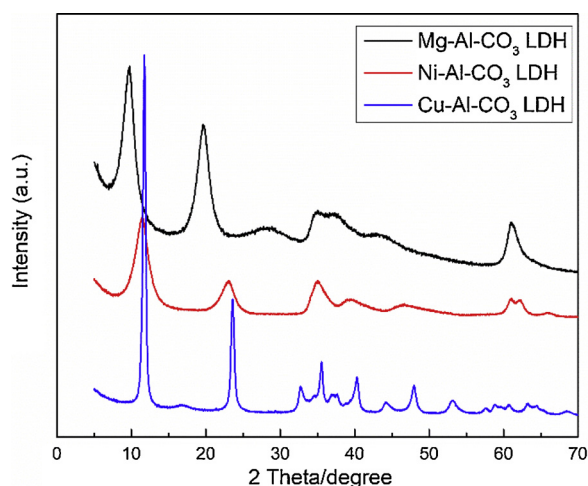


Fig. 2. XRD patterns of three fresh M (Mg/ Ni/ Cu)-Al-CO₃ LDHs synthesized by AMOST method.

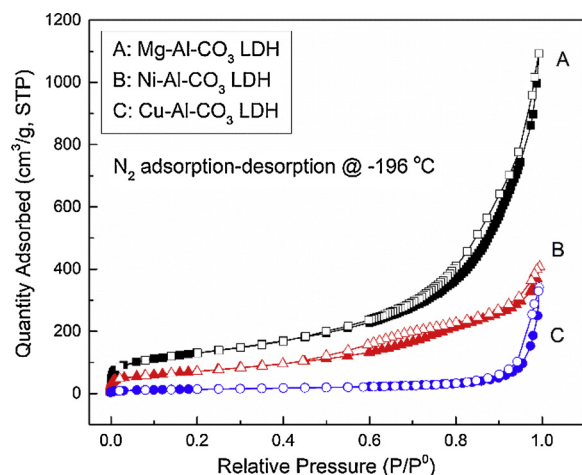


Fig. 3. N₂ adsorption-desorption isotherms of three M-Al-CO₃ LDHs at −196 °C.

rate of 800 rpm. Next, the sample was separated by centrifugation (5000 rpm) and washed with Milli-Q water till pH approached 7 and then washed with 500 ml of acetone and separated by centrifugation each time. After that, the obtained sample was dispersed in 300 ml of acetone with stirring overnight. The prolonged dispersion time during preparation would help LDH material increase the surface area. Finally, acetone was removed by rotary evaporator to recover the solid product Mg-Al-CO₃ LDH.

As for the synthesis of another two LDH samples Ni-Al-CO₃ and Cu-Al-CO₃, the similar procedure was followed with the only difference of using 37.5 mmol Ni(NO₃)₂·6H₂O and Cu(NO₃)₂·6H₂O respectively instead of Mg(NO₃)₂·6H₂O. The yields of these three LDHs can be seen in Table S-1.

2.3. Samples characterization

Powder X-ray diffraction (XRD) patterns were obtained using a Malvern Panalytical Xpert 3 X-ray diffractometer equipped with a Cu anode of 1.5406 Å λ radiation and the operating voltage was 40 kV. Diffraction patterns were recorded within the range of 2θ from 5° to 70° and the scan step size was 0.02°. The crystal structures of the newly synthesized M-Al-CO₃ LDH materials could be obtained by XRD results.

N₂ adsorption-desorption experiments at −196 °C were carried out using a Micromeritics 3-Flex. Prior to the analysis, all the three M-Al-CO₃ LDH samples were pre-activated at 120 °C under high vacuum for 12 h using a VacPrep 061 Sample Degas System, followed by in situ activation at 120 °C under high vacuum for 4 h using the Micromeritics 3Flex to thoroughly remove the pre-adsorbed impurities.

Transmission electron microscopy (TEM) was used to observe the morphology of the synthesized three M-Al-CO₃ LDH adsorbents with a Philips CM100 TEM. The fine powder of samples for characterization were dispersed in ethanol and sonicated for 10 min. Then a droplet of above solution was deposited on an ultrathin carbon film which was coated on a 300-mesh Cu grid.

Energy-dispersive X-ray spectroscopy (EDX) was used to determine the elemental composition of the three M (Mg, Ni, Cu)-Al-CO₃ LDH samples with an Aztec Energy X-MAX 50 EDS System.

Thermogravimetric analysis (TGA) was carried out to study the stability of the newly prepared LDH adsorbents with a METTLER TOLEDO TGA/DSC 1 instrument. The experiments were conducted under nitrogen atmosphere at a flow rate of 50 ml min^{−1}. To obtain the weight loss curves, the sample tested was kept at 25 °C for 10 min, prior to being heated from 25 to 900 °C at a ramping rate of 10 °C min^{−1}, and maintained at 900 °C for 10 min.

2.4. CO₂ and N₂ adsorption isotherms and cyclic CO₂ adsorption studies

CO₂ and N₂ adsorption isotherms were carried out using Micromeritics 3Flex. Before the measurements, all the samples were degassed at 180 °C for 4 h. The ramp rate for attaining the sample temperature was 5 °C/min. Cyclic CO₂ adsorption studies were carried out using the TGA instrument mentioned above. About 10 mg of sample was loaded in an alumina crucible and degassed in situ at 120 °C for 4 h under N₂ at a flow rate of 40 ml min^{−1}. After 4 h, when the mass became constant, N₂ was switched to CO₂ (99.999%, HKO Linde) and the weight uptake monitored at the same temperature for 90 min. Such cycle of regeneration and adsorption was repeated for 6 times to evaluate the cyclic CO₂ capacity.

3. Results and discussion

3.1. M (Mg/ Ni/ Cu)-Al-CO₃ Structure of M (Mg/ Ni/ Cu)-Al-CO₃ synthesized using AMOST method

3.1.1. XRD characterization of M-Al-CO₃ LDHs

As for the newly synthesized three M-Al-CO₃ LDHs using AMOST method, it is important to determine if they formed the structure of layered double hydroxide successfully. Fig. 2 shows the XRD patterns of these LDHs. All the three samples showed diffraction features of a typical hydrotalcite layered double hydroxide structure. The XRD patterns of all the three as-synthesized samples match well with the JCPDS reference patterns (Mg-Al-CO₃ LDH: JCPDS No: 00-014-0191, Ni-Al-CO₃ LDH: JCPDS No: 00-015-0087, Cu-Al-CO₃ LDH: JCPDS No: 00-037-0630), suggesting the pure phase LDHs were successfully obtained by

Table 1Surface and pore size characteristics of the synthesized M-Al-CO₃ LDH adsorbents calculated from N₂ adsorption isotherms at −196 °C.

Sample	Specific surface area* (m ² /g)	Micropore area [#] (m ² /g)	Mesopore area ^{\$} (m ² /g)	Total Pore Volume [@] (cc/g)	Micropore Volume [#] (cc/g)	Mesopore Volume ⁺ (cc/g)
Mg-Al-CO ₃	455.31	31.77	423.54	1.61	0.018	1.592
Ni-Al-CO ₃	249.45	19.45	230.00	0.59	0.009	0.581
Cu-Al-CO ₃	48.40	0.60	47.80	0.39	0.001	0.389

Calculations by * = BET, # = t-plot, \$ = Total area-micropore area, @ = single pore vol at p/p₀ = 0.99, + = total pore vol-micropore vol.

the modified method. Notably, for Mg-Al-CO₃ and Ni-Al-CO₃ LDHs, the peaks are broad and shifted toward lower 2θ in comparison with the Cu-Al-CO₃ LDH. This is indicative of larger basal spacing and smaller crystallite size in these two materials, suggesting the AMOST method has been successful in delaminating these two LDH materials. For Cu-Al-CO₃, the peak intensities are greatly increased over both 003 and 006 planes, suggesting a well crystallized Cu-Al-CO₃ LDH material was obtained. In other words, the AMOST method was less successful for the delamination of Cu-Al-CO₃ LDH than for the other two LDHs. The possible reason for this inadequate delamination of Cu-Al-CO₃ LDH could be as follows, unlike Ni²⁺ and Mg²⁺, which start precipitating as hydroxides with a pH above 9.2, the Cu²⁺ is known to get precipitated completely with a pH below 8 and its precipitation is highly accelerated with higher pH [37–39]. The synthesis pH of 10 thus would have led to rapid precipitation of Cu²⁺ and its incorporation in well stacked LDH structure within 0.5 h of aging before the acetone treatment. It has already been reported that the AMOST for delamination is more effective when a LDH is still being formed and stacking not occurred completely [34]. Even for Mg-Al-CO₃ LDH, if the material is aged and dried first, AMO solvent has little effect on dispersion [34]. Accordingly, in the case of rapidly crystallizing Cu-Al-CO₃ LDH, the acetone treatment would have led to an inadequate delamination and thus a better crystallinity. The estimated average crystallite size of three LDH samples were obtained from Scherrer equation using the most intense 003 plane [40]. The nanoplatelet size of Mg, Ni and Cu-Al-CO₃ LDH displayed 8.9, 9.0 and 19.0 nm, respectively.

3.1.2. Texture properties

Nitrogen sorption at −196 °C was carried out to investigate the porosity of the LDH adsorbents. Fig. 3 shows the adsorption and desorption isotherms of Mg-Al-CO₃, Ni-Al-CO₃ and Cu-Al-CO₃ LDH. As per IUPAC classification all the three materials possess a type III hysteresis loop indicative of typical mesoporosity [41,42]. BET surface area and pore volume calculated from these isotherms are given in Table 1. The previous reported Mg, Ni, Cu-CO₃-LDH materials synthesized by co-precipitation method usually showed quite low surface area around 14–110 m²/g [3,43,44]. As for our LDH materials synthesized with AMOST method, they showed largely increased surface area, 3–17 times than those with co-precipitation method. Specially, Mg-Al-CO₃ and Ni-Al-CO₃ LDHs showed 455.31 and 249.45 m²/g, respectively, much higher than that of Cu-Al-CO₃ (48.40 m²/g). Surface area results along with the XRD results point out that materials with lower crystallinity show high surface area and vice versa.

Barrett-Joyner-Halenda (BJH) method was used to study the pore size distribution of three M-Al-CO₃ LDH samples (in Fig. S1). All the materials showed in general a wide pore size distribution which was as anticipated because of the slit shaped pore spaces of varying diameter formed between the randomly oriented nanoplatelets of these materials.

3.2. Morphology and elements composition

TEM characterization provided the morphology for our three LDHs. Fig. 4 shows that all the three AMOST synthesized M-Al-CO₃ LDH are all composed of thin and delaminated 2D nanoplatelets but with big

difference in particle size. For Mg-Al-CO₃ and Cu-Al-CO₃ LDHs, they showed nano-platelet-like morphology with poor dispersion. But for Ni-Al-CO₃ LDH, a nano-flower-like morphology and uniform dispersion were presented, in agreement with previous report [36]. The nano-flower morphology allows for the constituent nano-particles to interleave with each other to easily form a porous network [45], which is likely to present minimal gas diffusion limitation and thus improve adsorption kinetics. Even though the three LDH materials were synthesized using the same preparation methods, they showed different particle sizes and morphology. It may be attributed to different divalent metal LDHs need different aging times and different organic solvent amounts during the synthesis. Further work may focus on optimizing the preparation of different divalent metal LDHs to provide desired morphology for different LDH samples.

EDX was used to determine the elemental composition of the LDH adsorbents. The peaks for Mg, Ni, Cu, and Al in Fig. 5 show the divalent and trivalent elements composition in the LDHs. The formula of different LDH samples are shown in Table S-2. The experimental atomic ratio between divalent cation (Mg, Ni, Cu) and trivalent cation (Al) varied from 1.2 to 1.9, which was lower than 3, the ratio taken in synthesis mixture. The possible underlying reason is the adoption of shortened aging time (0.5 h) in our preparation process, resulting in the leaching of unreacted cations and the resultant lower M/Al atomic ratio.

3.3. Stability of M (Mg/ Ni/ Cu)-Al-CO₃ synthesized with AMOST method

It is well known that the thermal stability is vital for adsorbents especially for those working at high temperatures. TGA and XRD at different temperatures were performed to evaluate the stability of the as-synthesized M-Al-CO₃ LDH samples. As shown in Fig. 6, all the three LDHs showed a slight weight loss below 120 °C owing to the desorption of acetone and water. Compared with Mg-Al-CO₃ and Ni-Al-CO₃ LDH, Cu-Al-CO₃ LDH presented a lower weight loss below 120 °C, which is probably due to the less amount of acetone and water component in Cu-Al-CO₃ LDH. Subsequently, for Mg-Al-CO₃ LDH, the sharp weight loss started above 120 °C and finally reached a stable weight around 400 °C. This is attributed to the collapse of the layered hydroxide structure to Mg-Al mixed oxide structure via decomposition and release of the interlayer carbonate as CO₂ [25]. The XRD peaks (Fig. S2(a)) of the calcined Mg-Al-CO₃ at 400 °C indicated the formation of Mg-Al mixed oxide, corresponding to the poorly crystalline MgO dispersed in amorphous alumina. The phase segregation of MgO from the mixed oxide occurred at 750 °C, as evidenced by more intense MgO peaks indicative of the higher crystallinity.

Ni-Al-CO₃ LDH layered structure was less thermally stable than the Mg-Al-CO₃ LDH structure. From the XRD of Ni-Al-CO₃ LDH (Fig. S2(b)), the disappearance of 006 plane after thermal treatment at 250 °C indicated the layered structure partially collapsed. The LDH structure of Ni-Al-CO₃ completely disappeared above 400 °C and formed poorly crystalline NiO on alumina. With further increase in temperature to 750 °C, NiO phase segregation from mixed oxide occurred and crystallite size increased suggested by the intensity of respective XRD peaks. Cu-Al-CO₃ LDH began to decompose below 250 °C, showing the least thermal stability. Upon decomposition, the Cu-Al mixed oxide formed

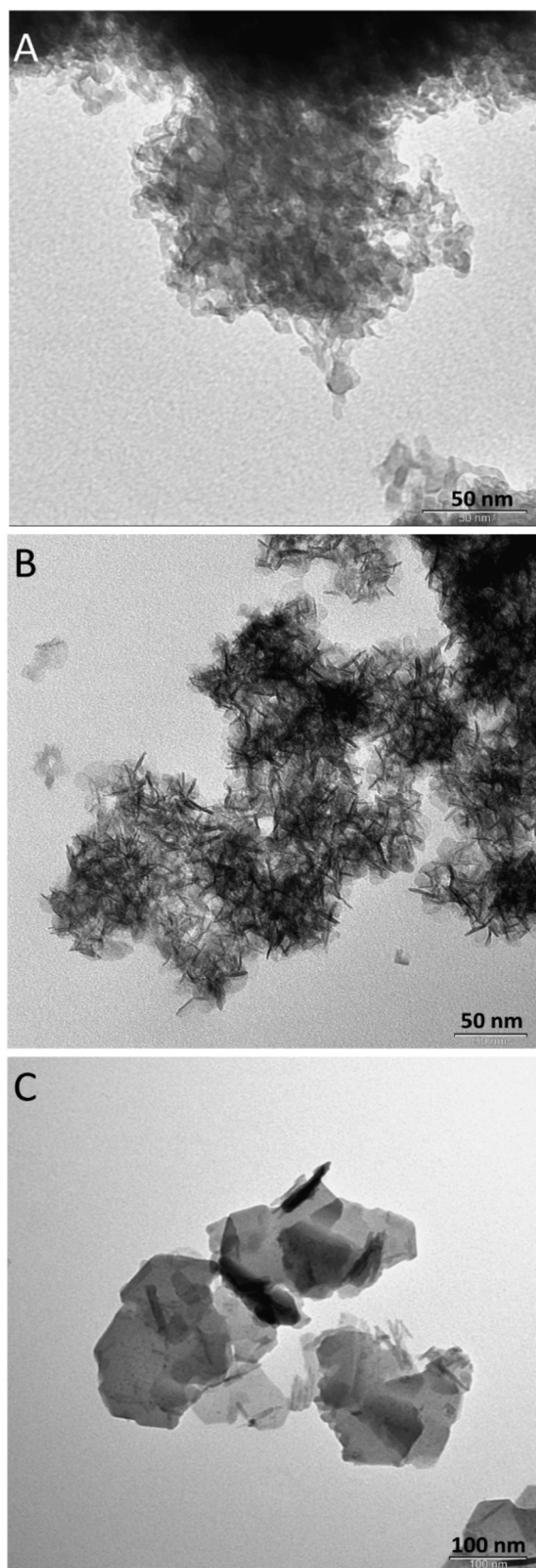


Fig. 4. TEM images of M-Al-CO₃ LDHs synthesized by AMOST: (A) Mg-Al-CO₃, (B) Ni-Al-CO₃ and (C) Cu-Al-CO₃.

and the phase segregation of CuO occurred subsequently with the increasing calcination temperature. The crystallinity of CuO continued to increase with the temperature reaching maximum at 750 °C (Fig. S2(c)). From the combined XRD pattern and TGA results, we concluded that Mg-Al-CO₃ and Ni-Al-CO₃ LDHs showed the relatively high thermostability compared with Cu-Al-CO₃ LDH.

3.4. CO₂ capture performance

3.4.1. CO₂ adsorption capacity

CO₂ adsorption capacity at 50 °C for all three adsorbents is given in Fig. 7 (a). The CO₂ uptake of all three LDH samples increased with the increase in absolute pressure. For the Mg-Al-CO₃ LDH, the CO₂ adsorption amount reached up to 0.83 mmol/g (1200 mbar), which is higher than the previous reported ones (0.28–0.6 mmol/g) using coprecipitation method [3,46]. Ni-Al-CO₃ LDH showed a slightly higher CO₂ adsorption capacity of 0.87 mmol/g than Mg-Al-CO₃ LDH. For Cu-Al-CO₃, the CO₂ capture capacity is much lower than the other two, only achieving 0.21 mmol/g at the pressure of 1200 mbar. Despite the lower surface area of Ni-Al-CO₃ LDH than Mg-Al-CO₃ LDH, the CO₂ capacity of Ni-Al-CO₃ LDH (0.87 mmol/g) is slightly higher than that of the Mg-Al-CO₃ LDH (0.83 mmol/g). Notably, the exposed surface area is not the only factor determining the CO₂ capacity but the number of exposed active basic adsorption sites governs the CO₂ adsorption capacity. As confirmed by EDX (Table S2) the Ni/Al ratio in Ni-Al-CO₃ LDH is 1.9 against 1.2 for Mg/Al ratio in Mg-Al-CO₃ LDH. Owing to higher Ni/Al ratio in Ni-Al-CO₃ LDH, the number of exposed basic active sites could be higher even though the surface area is lower than Mg-Al-CO₃ LDH.

In order to test the effect of temperature on CO₂ adsorption capacity of the adsorbents, the adsorption experiments were carried out at 80 °C and 120 °C as these temperatures cover the whole temperature range applicable to post combustion CO₂ capture. The CO₂ adsorption isotherms of the adsorbents are shown in Fig. 7(b). Interestingly, the CO₂ capacities of three M-Al-CO₃ LDHs at 80 °C were almost the same as that at 50 °C. The CO₂ uptake did not decrease with the increased temperature, contradicting to the physisorption feature. Therefore, we contended that both physisorption as well as chemisorption occurred on these M-Al-CO₃ LDHs.

With further increase of the testing temperature to 120 °C, the Ni-Al-CO₃ LDH still showed the highest CO₂ capture ability among the three adsorbents (Fig. 7(c)). At 1200 mbar, the uptake amounts of CO₂ were 0.60, 0.35 and 0.10 mmol/g for Ni, Mg, and Cu-Al-CO₃ LDHs, respectively. The CO₂ uptake of all the investigated LDH materials decreased at 120 °C as compared to that at 80 °C. In particular, the Mg-Al-CO₃ and Cu-Al-CO₃ LDHs showed pronounced decrease of 58% and 64% (Fig. 7(d)). Even though the CO₂ capacity of Ni-Al-CO₃ decreased 28%, it was still higher than the previously reported CO₂ adsorption capacity range of 0.18–0.5 mmol/g for LDHs [46]. The decrease of CO₂ adsorption capacity for all the three LDHs from 80 °C to 120 °C, is probably due to the nature of physisorption dominated the change in the trend of uptake within this temperature range, the higher temperature the lower the adsorption capacity.

3.4.2. N₂ adsorption capacity

As we all know, flue gas is a mixture of many gas components and the two main components are carbon dioxide and nitrogen, which account for approximately 90–96% [47]. The general molar ratio used for simulated flue gas is CO₂ : N₂ (85:15) [48]. A promising CO₂ capture adsorbent should have both high capacity for CO₂ and low capacity for N₂. In order to access the adsorption selectivity of CO₂ over N₂, we further measured the single component adsorption of N₂ using 3-Flex on the three LDH samples at 50, 80, and 120 °C, respectively. As can be seen in Fig. 8, at 50 °C, Mg-Al-CO₃ and Ni-Al-CO₃ LDHs showed a similar capacity for N₂ of 0.02 mmol/g at 1200 mbar, which was a little higher than that of Cu-Al-CO₃ LDH, below 0.01 mmol/g, at the same

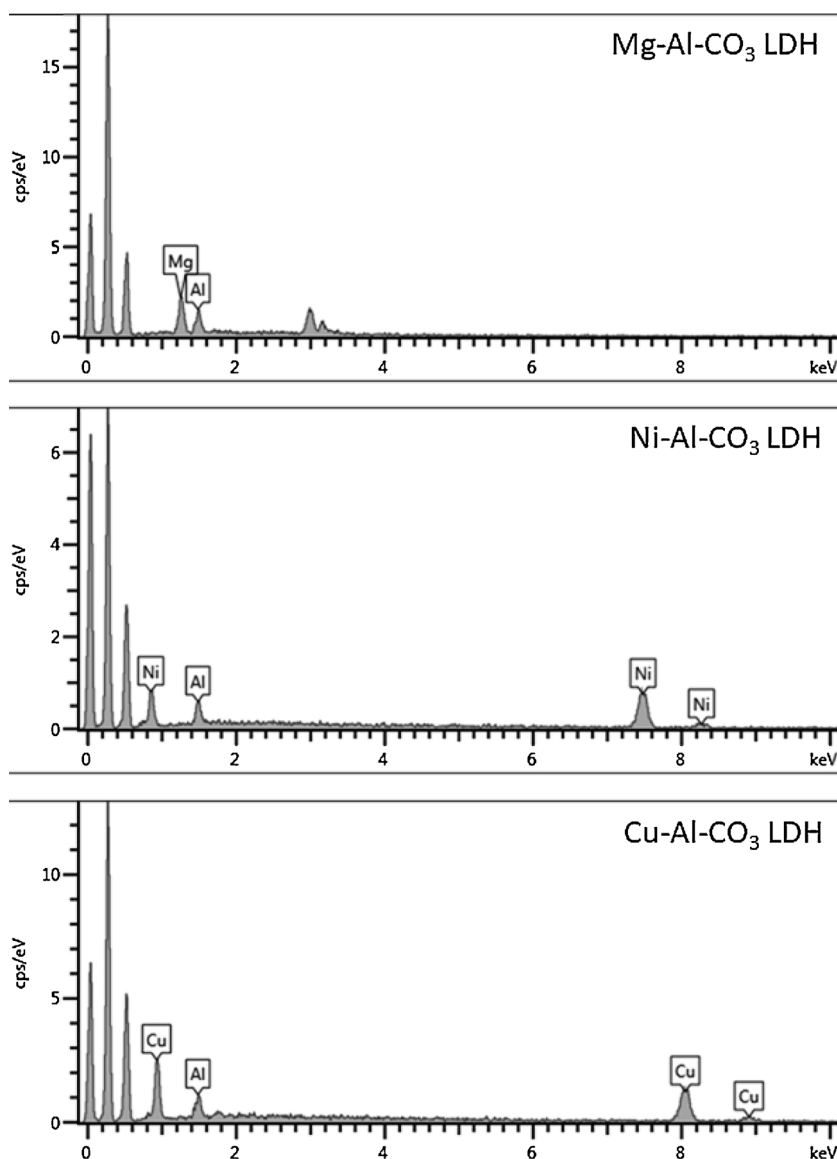


Fig. 5. EDX of three M-Al-CO₃ LDHs synthesized by AMOST.

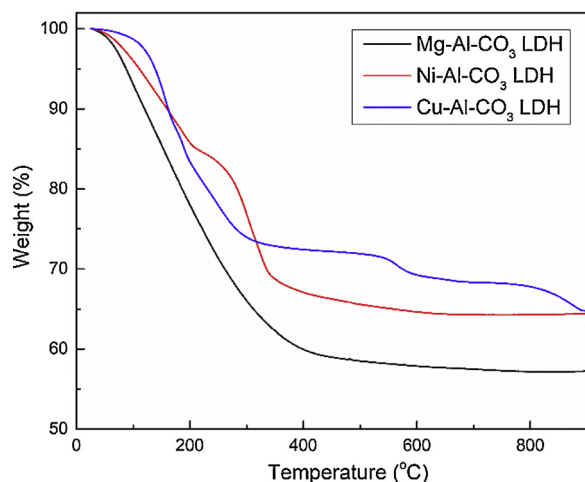


Fig. 6. TGA curves of three M-Al-CO₃ LDHs synthesized by AMOST.

pressure. The reason maybe is Mg-Al-CO₃ and Ni-Al-CO₃ LDHs have larger surface area than Cu-Al-CO₃ LDH. Note that all the three samples show non-smooth N₂ adsorption isotherms. The non-smooth N₂ adsorption isotherms of three LDHs at 50 °C suggest that, the N₂ adsorption may not truly reach the thermodynamic equilibrium. Note that we keep the default equilibrium setting on our gas adsorption analyzer for the sake of fair comparison of different materials. At this temperature, the recorded N₂ adsorption capacity should be lower than the equilibrium capacity due to the restricted mass transfer. When the temperature was increased to 80 °C, Ni-Al-CO₃ LDH showed the lowest N₂ uptake below 0.07 mmol/g, which was almost half than that of Mg-Al-CO₃. However, we observed that the N₂ adsorption amount of all the three LDHs at 80 °C was higher than that at 50 °C. This is because the elevated temperature alleviated the mass transfer limitation, leading to increased N₂ uptake on all the three LDHs. When the temperature was further increased to 120 °C, for Ni-Al-CO₃ and Cu-Al-CO₃ LDH, the restricted mass transfer no longer governed the gas uptake while the nature of physisorption dominated the change in the trend of uptake, i.e., the higher temperature the lower adsorption capacity. However, the N₂ adsorption of Mg-Al-CO₃ LDH still did not reach the thermodynamic equilibrium, therefore it displayed an increased N₂ uptake with elevated temperature.

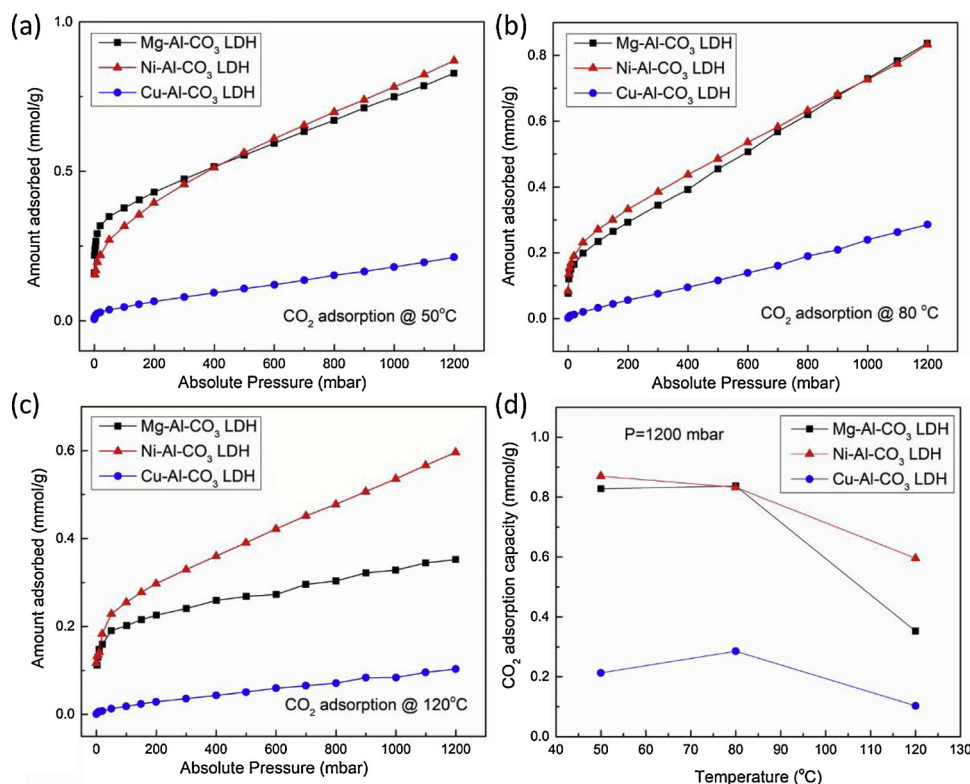


Fig. 7. CO₂ adsorption amount for three M-Al-CO₃ LDHs at three temperatures, 50 °C (a), 80 °C (b), and 120 °C (c), and the CO₂ adsorption capacity at 1200 mbar (d).

3.4.3. The ideal CO₂/N₂ selectivity for flue gas

To evaluate the potential separation ability, the ideal selectivities of three LDHs for CO₂ over N₂ were calculated using the equation [49] for binary gas mixture as follows:

$$S = \frac{x_1/x_2}{y_1/y_2}$$

where x denotes adsorption amount and y denotes gas phase pressure, and 1 and 2 the strongly and weakly adsorbed components, respectively, of the binary gas mixture. Besides, the most common molar ratio in simulated flue gas for CO₂ : N₂ (85:15) was used, and thus here $y_1 = 0.15$ and $y_2 = 0.85$.

The calculated ideal selectivities of CO₂/N₂ are displayed in Table 2. At 50 °C, all the three LDH adsorbents showed high CO₂/N₂ selectivity with the highest achieved on Ni-Al-CO₃, i.e., 166. With the increase in adsorption temperature to 80 °C, all three samples showed the declined selectivity of CO₂/N₂, with Ni-Al-CO₃ LDH showing the highest selectivity of 53, followed by Mg-Al-CO₃ and Cu-Al-CO₃ (both below 17). Interestingly, the selectivity for Ni-Al-CO₃ LDH at 120 °C was maintained at a high level, i.e., 58, which is 6–10 times higher than that of the other two adsorbents. In general, the newly synthesized Ni-Al-CO₃ LDH with AMOST method shows the highest CO₂/N₂ selectivity at all the three tested temperatures and is the best adsorbent for the CO₂ capture from flue gas.

The comparison among three M-Al-CO₃ LDH adsorbents shows that, Ni-Al-CO₃ LDH is the most promising one because it has both high CO₂ adsorption capacity and high CO₂/N₂ selectivity. The adsorption isotherms suggest that the highest CO₂ adsorption capacity could reach 0.87 mmol/g and the CO₂/N₂ selectivity could achieve 166 at 50 °C at 1200 mbar. We contend that the reason why the newly synthesized Ni-Al-CO₃ LDH possessing good CO₂ capture ability is that both the physisorption and chemisorption occurred between CO₂ and adsorbent, although the surface area and pore volume of Ni-Al-CO₃ LDH was not the highest among the three LDHs. Besides, Ni-Al-CO₃ LDH showed a similar CO₂ uptake when the temperature increased from 50 °C to 80 °C.

The CO₂ uptake did not decrease with the increased temperature, contradicting to the physisorption feature. Therefore, we believe that there should be some active adsorption sites between divalent cation Ni²⁺ and CO₂, resulting in the increased interactions between CO₂ and Ni-Al-CO₃ LDH contributing to chemisorption. Future works will focus on studying the interactions between the highly efficient Ni-Al-CO₃ LDH and CO₂ molecule to reveal the adsorption mechanisms.

As previously reported [3], it is widely accepted that Mg LDH-derived materials, the mixed metal oxides calcined above 400 °C, usually present the best CO₂ capture performance ranging from 0.28–0.72 mmol/g [3]. At the same time, it is generally acknowledged that a fresh LDH has no CO₂ capture capacity. Only upon thermal treatment, LDHs gradually lose the interlayer water and carbonate, and then dehydroxylate and decarbonate to a large degree, leading to the formation of MMO. The high CO₂ capacity of Mg LDH derived material relied on active Mg-O adsorption sites [3]. In our study, the fresh Ni-Al-CO₃ LDH synthesized by AMOST method without any calcination at high temperatures showed a superior CO₂ adsorption capacity and CO₂/N₂ selectivity even at high temperature (120 °C). The synthesis method by using AMOST for LDH materials could provide a new route to develop LDHs for CO₂ capture and other CO₂ related catalytic reactions.

3.4.4. Cyclic performance of Ni-Al-CO₃ LDH for CO₂ capture

We further studied the reusability of the best performing adsorbent identified in this study, Ni-Al-CO₃ LDH, as representative CO₂ capture adsorbents using TGA. The cyclic CO₂ adsorption results are shown in Fig. 9. As anticipated, the initial (Cycle-1) CO₂ adsorption amount at 1 atm and 120 °C on Ni-Al-CO₃ LDH was 0.46 mmol/g, which was slightly lower than that obtained from static CO₂ adsorption at 1013 mbar (0.54 mmol/g) using 3 Flex. This is a common phenomenon for dynamic gas adsorption compared with static gas adsorption process. Because the gas adsorption amount of adsorbents achieved at dynamic conditions is the result of gas adsorbates achieving local equilibrium with continuous gas flowing, which is generally lower than the gas uptake under multiple static equilibrium after complete

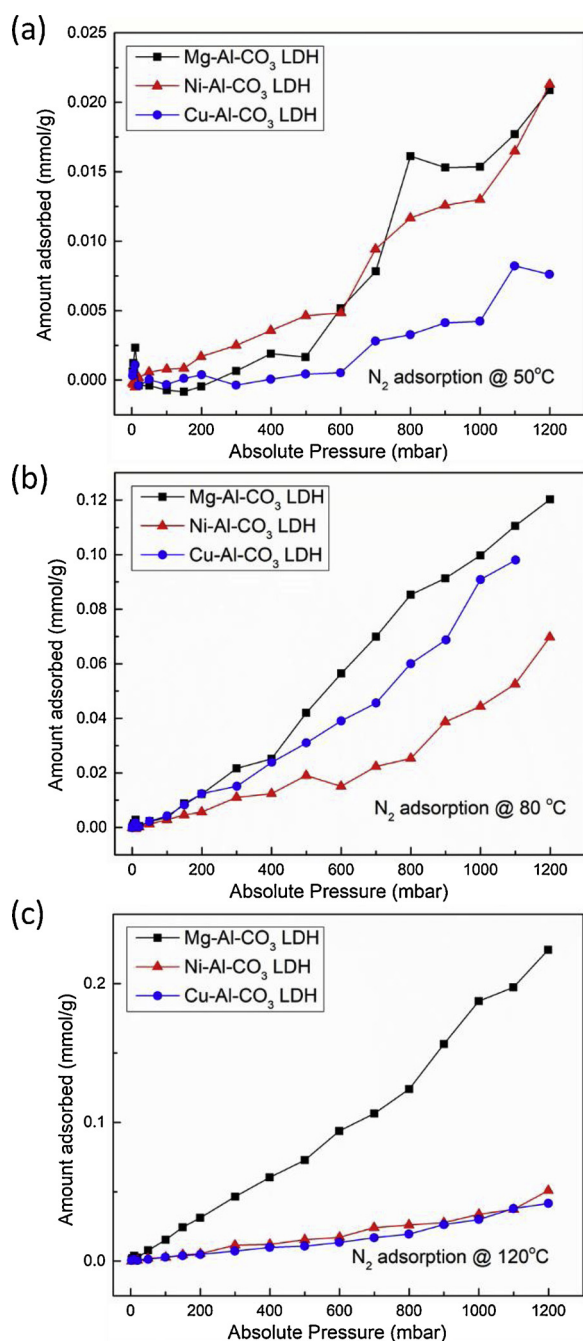


Fig. 8. N₂ adsorption amount for three M-Al-CO₃ LDHs at three temperatures, 50 °C (a), 80 °C (b) and 120 °C (c).

Table 2
Selectivity of M-Al-CO₃ LDH adsorbents for CO₂/N₂ at different temperatures.

Adsorbent	Selectivity of CO ₂ /N ₂		
	50 °C	80 °C	120 °C
Mg-Al-CO ₃	146	17	9
Ni-Al-CO ₃	166	53	58
Cu-Al-CO ₃	85	4	6

saturation [50,51]. After the regeneration of the adsorbent, we carried out another five consecutive runs. The CO₂ capture performance of Ni-Al-CO₃ LDH displayed no significant change during the five adsorption-regeneration cycles. Notably, the last two cycle experiments showed the

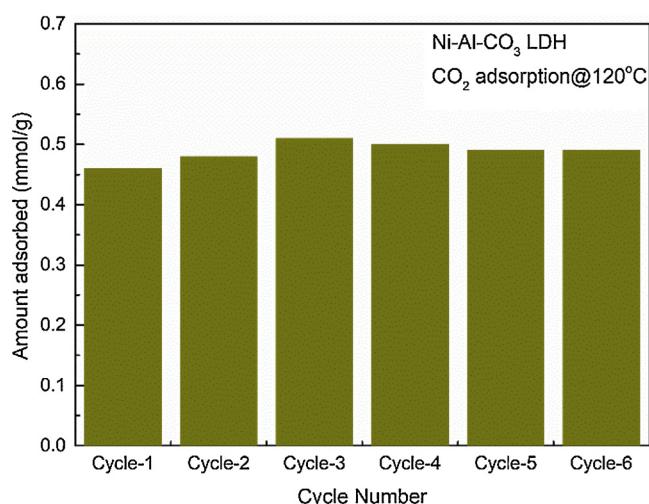


Fig. 9. CO₂ adsorption capacity for Ni-Al-CO₃ LDH at 120 °C and ambient pressure over multiple cycles of adsorption-regeneration process.

same CO₂ adsorption amount of 0.49 mmol/g, indicating the good stability of the newly synthesized Ni-Al-CO₃ LDH with AMOST method as a CO₂ capture adsorbent.

4. Conclusion

We synthesized three M (Mg, Ni and Cu)-Al-CO₃ layered double hydroxides using aqueous miscible organic solvent treatment (AMOST) method and studied their potential for CO₂ capture. The newly prepared Ni-Al-CO₃ LDH displayed a high surface area at 249.45 m²/g and uniform nano-flower-like morphology. Of the three adsorbents, Ni-Al-CO₃ LDH showed the highest CO₂ adsorption capacity of 0.87 mmol/g and CO₂/N₂ selectivity of 166 at 50 °C. The superior CO₂ adsorption capacity and selectivity of Ni-Al-CO₃ LDH may result from the chemisorption of CO₂ onto Ni sites. Multiple cyclic adsorption-regeneration experiment results indicated that Ni-Al-CO₃ LDH has good stability as CO₂ capture adsorbent. This is the first report of un-calcined Ni-Al-CO₃ LDH adsorbent capable of exhibiting superior CO₂ capture performance to the well-reported optimal Mg LDH which has long been deemed to be the best LDH-based CO₂ adsorbent. The synthesis method by using aqueous miscible organic solvent treatment for LDH materials could provide a new route to develop LDHs for a wide range of applications such as in CO₂ capture and CO₂ catalytic reactions.

Declarations of interest

None.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhazmat.2019.03.077>.

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